

Conceptual Design Of a Quadrotor

□ Constant Parameters Used in the Model

Parameter	Symbol (ρ)	Value	Description
Air density		1.225 kg/m ³	Standard sea-level air density
Profile drag coefficient	(C)	0.01	Blade profile drag coefficient
Induced power factor	(κ)	1.15	Accounts for wake rotation and non-ideal losses
Motor efficiency	(η)	0.8	Electrical to mechanical conversion efficiency
Number of blade radial segments	(n)	20	Used for numerical integration in BEMT

□ Mission parameters

Parameter	Value	Description
Payload mass	300 g	Required payload capacity
Endurance	20 minutes	Required flight duration
Battery configuration	Variable	Number of Li-Po cells
Voltage per cell	3.7 V	Nominal voltage of Li-Po battery cell

The total battery voltage was calculated as

$$V_{total} = N_{cells} \times 3.7$$

where N_{cells} represents the number of battery cells connected in series.

To ensure sufficient thrust margin for maneuverability, the maximum thrust requirement was assumed to be twice the hover thrust, i.e.

$$T_{max} = 2T_{hover}$$

□ Initial Rotor and Battery Specifications

For the first baseline model, the following rotor geometry parameters were assumed.

Parameter	Value
Root pitch angle (θ_{root})	22°
Blade twist	17° per blade radius
Rotor radius (R)	0.0979 m
Blade chord (c)	0.0163 m
Root cutout (R_{cut})	0
Number of blades (N_b)	2

The blade pitch distribution was modeled as

$$\theta(r) = \theta_{root} - 17r$$

where r represents the non-dimensional radial position along the blade.

The aerodynamic coefficients of the rotor were computed using BEMT. The resulting coefficients were:

$$C_T = \text{Rotor } C_t: 0.00832763$$

$$C_p = \text{Rotor } C_p: 0.00070640$$

These coefficients were then used to determine rotor speed and power requirements using the relations

$$\begin{aligned} T &= C_{TQ} (A \Omega)^2 R \\ P &= C_{PQ} (A \Omega)^3 R \end{aligned}$$

where

$$A = \pi R^2$$

is the rotor disk area.

□GTOW Convergence Using Iterative Sizing Algorithm

The quadrotor sizing algorithm was executed using the configuration described above.

The initial gross take-off weight (GTOW) was estimated as three times the payload mass, as recommended in the assignment guidelines.

$$GTOW_{initial} = 3 \times \text{payload}$$

The algorithm iteratively updated the GTOW by computing:

1. Required rotor thrust
2. Rotor angular velocity
3. Rotor power consumption
4. Battery capacity required for the mission
5. Motor, ESC, battery, rotor and airframe weights
6. Electronics (FC, Sensors, Wiring) weight is assumed to be ~15g

The updated GTOW was calculated as

$$GTOW = 4(m_R + m_{\text{motor}} + m_{\text{ESC}}) + m_{\text{battery}} + m_{\text{airframe}} + m_{\text{electronics}} + \text{payload}$$

less than

The algorithm was repeated until the difference between consecutive GTOW estimates was less than 5g.

The sizing algorithm converged after 9 iterations.

```
***** Iteration 9 *****

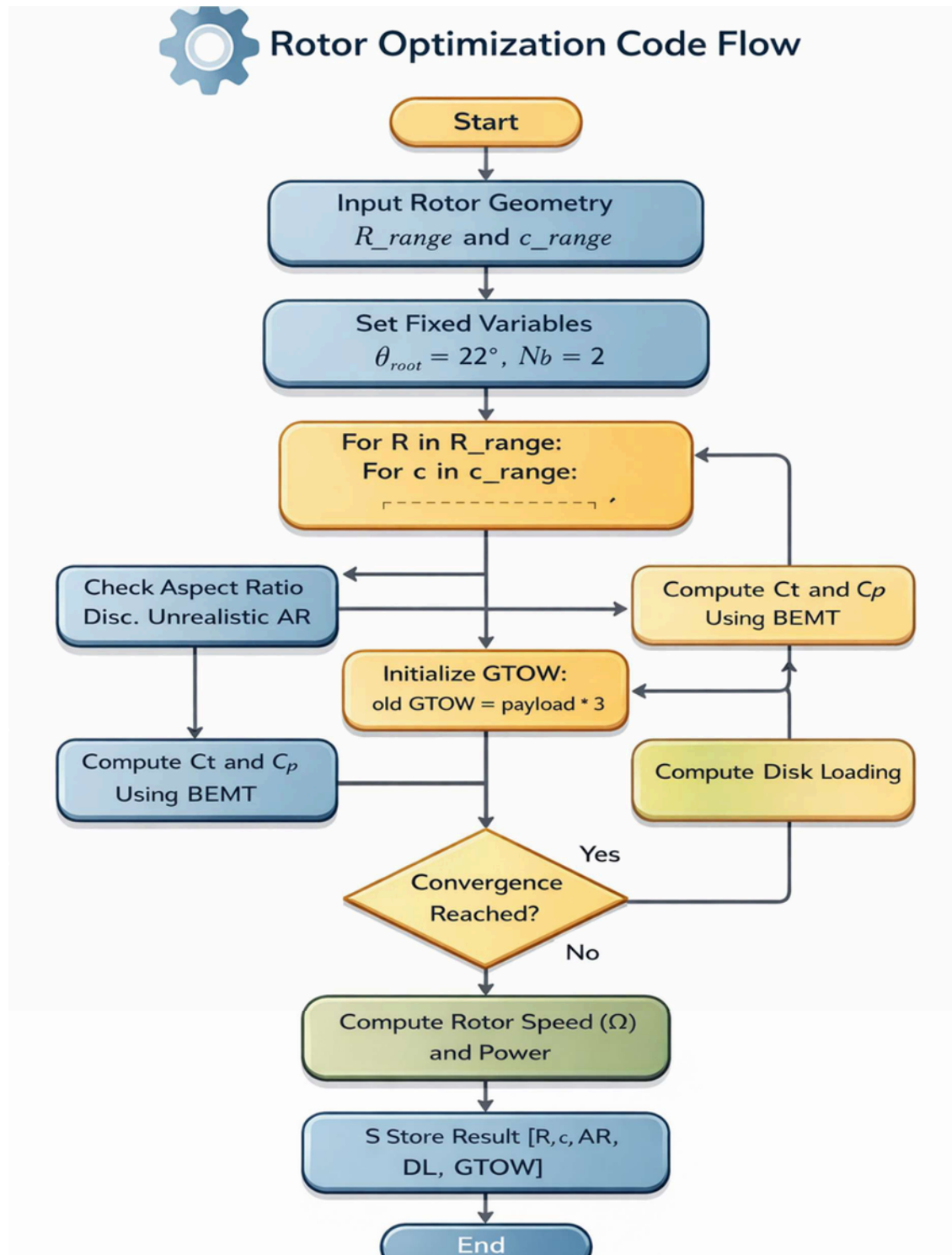
Omega for hover (RPM):      9179.84
Omega maximum (RPM):      12982.25
Hover thrust per rotor (N):  2.72
Max thrust per rotor (N):   5.44
Hover power per rotor (W):   27.15
Max power per rotor (W):    76.79

Max Current drawn per motor (A):  5.19
Kv of each motor (RPM/V):      877.18
Mission energy (J) or (Ws):    162892.25
Battery cell capacity (mAh):    3057.29

GTOW = 1113.55, Rotor = 5.20, Motor = 25.65, ESC = 3.97, Battery = 329.35, Airframe = 344.89
```

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The designed rotor (~6.7 – 7.7 inch diameter, two-blade configuration) aligns well with commercially available propellers used in quadrotors in the 1–1.5 kg weight range. Furthermore, the selected 4S Li-Po battery configuration is standard for such platforms, offering a suitable trade-off between power delivery and efficiency.



Parameter ranges to Test :

```
theta_root_range = np.linspace(20,30,50)
# 20° to 30° at 0.2° intervals
disk_loading_range = np.linspace(40,90,100)
# 40 to 90 N/m2 at 0.5 intervals
aspect_ratio_range = np.linspace(3,13,50)
# 3 to 13 at 0.2 intervals
```

```
Nb_range = np.array([2,3]) # Discrete values of 2 & 3
```

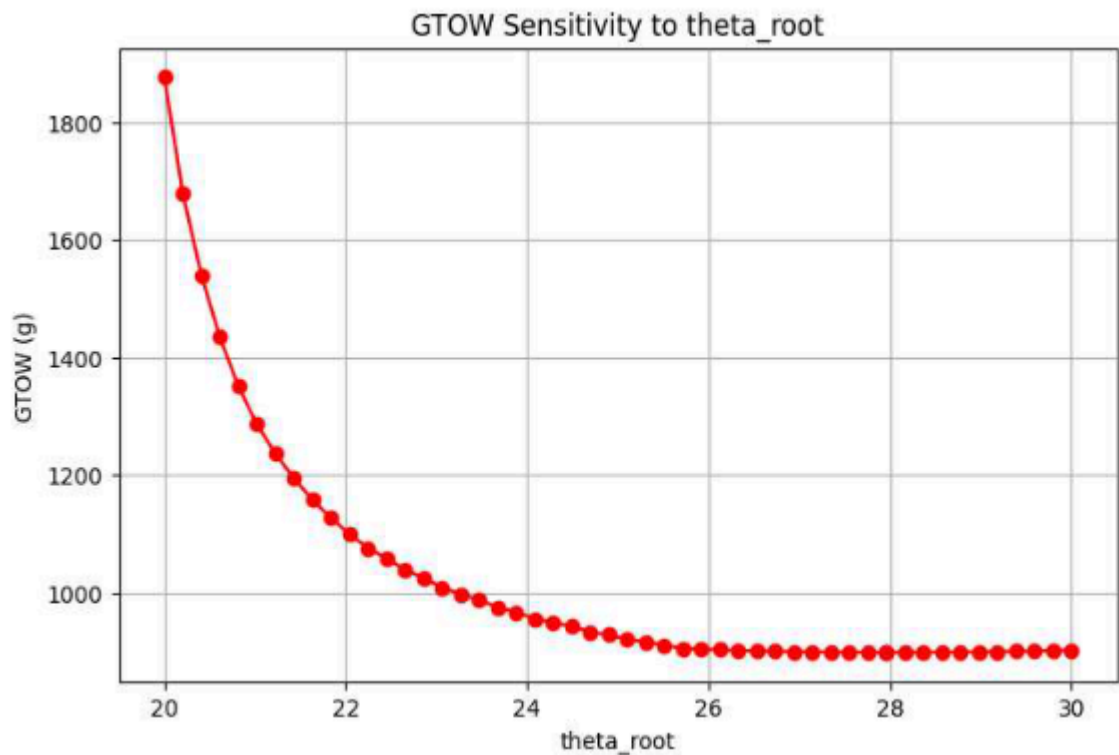
```
cells_range = np.arange(3,7) # Discrete values of 3, 4, 5, 6
```

Execute the Sensitivity Plot function :-

1. GTOW Sensitivity vs theta_root

Optimal theta_root: 27.95918367346939

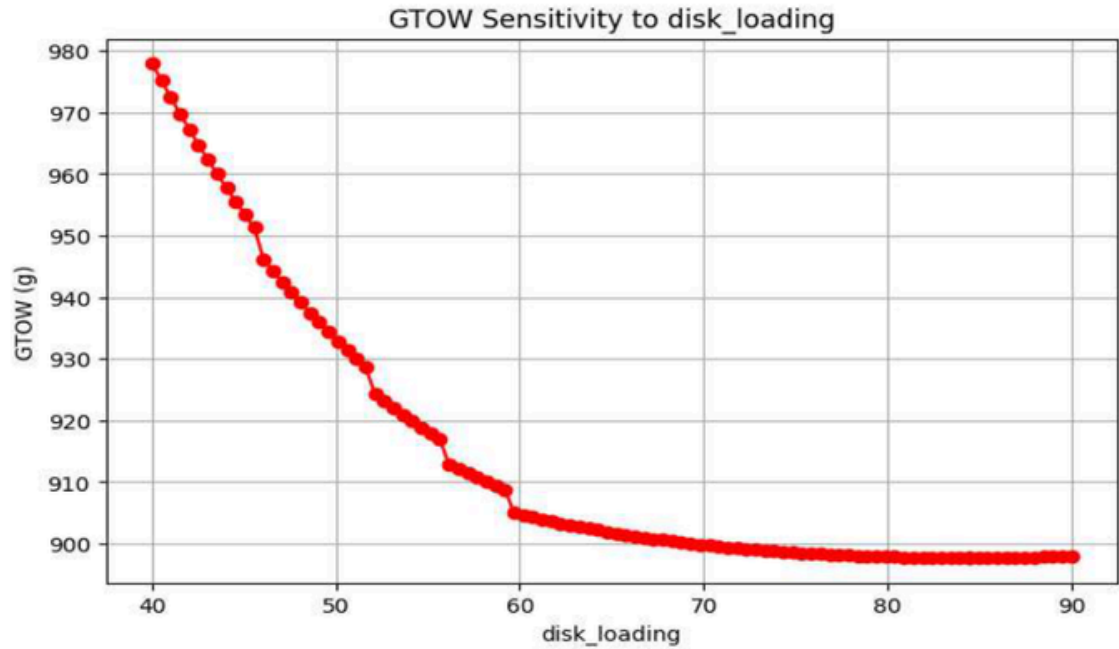
Minimum GTOW: 897.66 g



2. GTOW Sensitivity vs disk_loading

Optimal disk_loading: 84.44444444444446

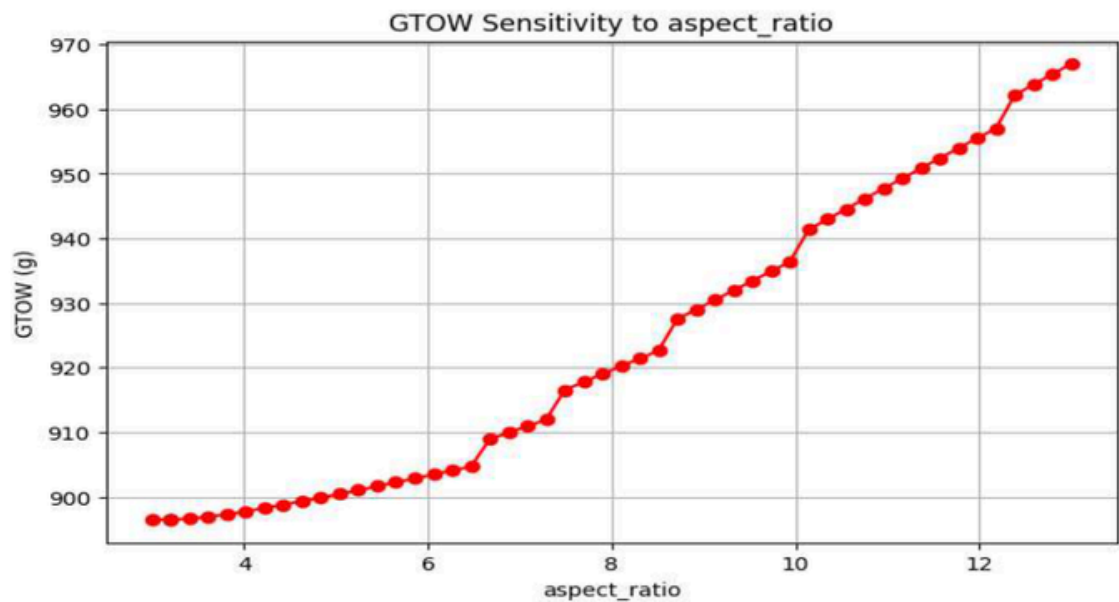
Minimum GTOW: 897.70 g



3. GTOW Sensitivity vs aspect_ratio

Optimal aspect_ratio: 3.204081632653061

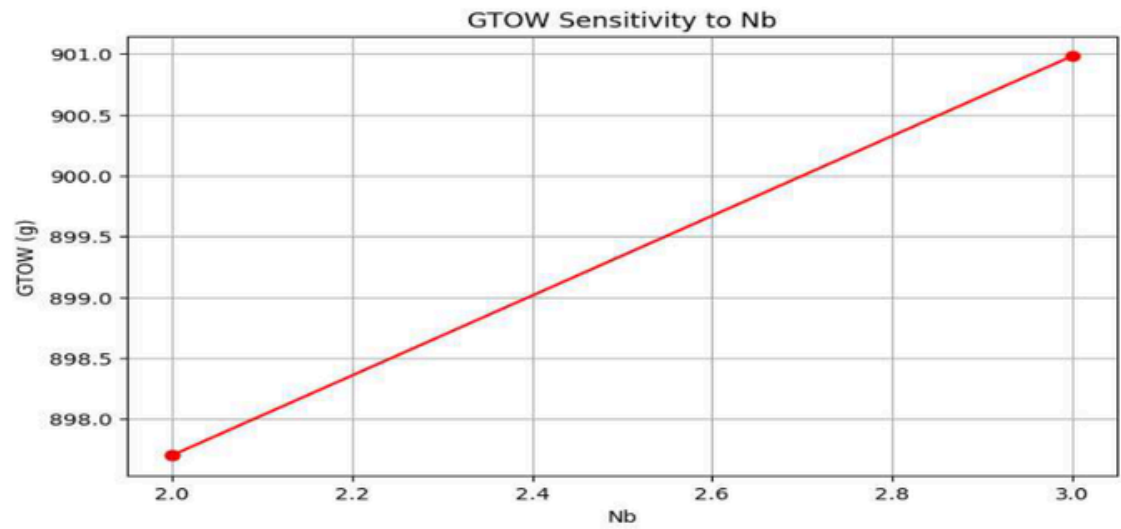
Minimum GTOW: 896.46 g



4. GTOW Sensitivity vs nos_of_blades

Optimal Nb: 2

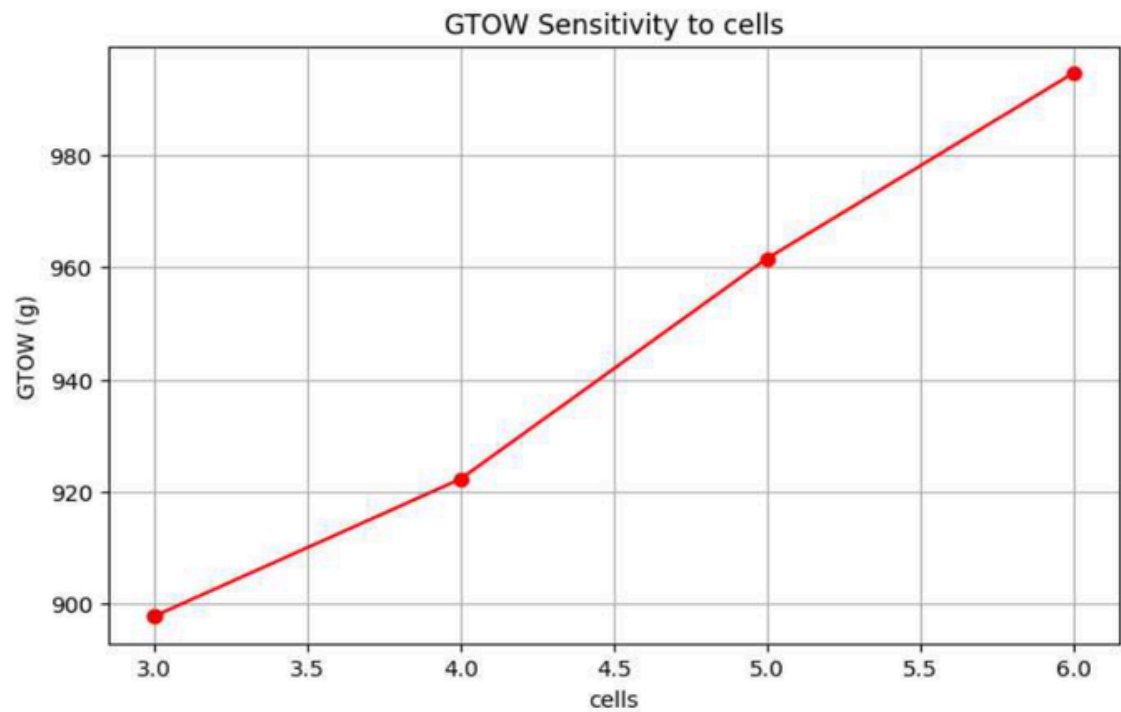
Minimum GTOW: 897.70 g



5. GTOW Sensitivity vs nos_of_cells

Optimal cells: 3

Minimum GTOW: 897.70 g



To get optimal GTOW value, we iterate over all the parameter ranges and run the sizing algorithm using all combinations. However, the intervals were reduced here to conserve computational expense. Below are the results obtained:

```
[48]
...
Best Design Found
GTOW: 897.701848131753 g
theta_root: 27.7777777777778
disk_loading: 84.44444444444444
aspect_ratio: 4.0
Nb: 2
cells: 3
```

Running the sizing code using these parameters again to get details of the quadrotor:

```
...
***** Iteration 0 *****

Omega for hover (RPM):      6014.42
Omega maximum (RPM):       8505.68
Hover thrust per rotor (N): 2.21
Max thrust per rotor (N):   4.41
Hover power per rotor (W):  17.78
Max power per rotor (W):    50.30

Max Current drawn per motor (A): 4.53
Kv of each motor (RPM/V):  766.28
Mission energy (J) or (Ws): 106702.54
Battery cell capacity (mAh): 2670.23

GTOW = 904.65, Rotor = 5.20, Motor = 18.54, ESC = 3.48, Battery = 213.22, Airframe = 282.53

Rotor Radius (m):          0.09790
Rotor Chord (m):           0.01630
Rotor Thrust Coefficient (Ct): 0.01573945
Rotor Power Coefficient (Cp): 0.00164530

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Add Code Cell
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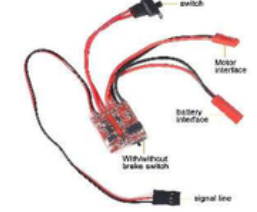
Comparative Analysis

A comparative analysis was performed between the initial baseline design and the optimized configuration obtained from sensitivity analysis.

- The optimized configuration achieves a significant reduction in GTOW (~230 g lighter) compared to the initial baseline design.
- The optimized design operates at a higher disk loading (~84 N/m² vs ~65–70 N/m²), resulting in smaller rotor size and reduced structural weight.
- The root pitch angle is higher (~28° vs 22°), which increases thrust coefficient and allows operation at lower rotor speeds.
The aspect ratio is reduced (~3–4 vs ~6), leading to a higher chord length and increased solidity.
- Due to higher solidity, the rotor produces required thrust at lower RPM, reducing induced power requirements.
- The number of blades remains unchanged ($N_b = 2$), which is consistent with standard quadrotor configurations.
- The optimized design suggests a 3-cell battery configuration (vs 4-cell in baseline), reducing overall battery weight.
The reduction in battery requirement directly contributes to a lower airframe weight, as airframe mass scales with battery mass.
- The optimized configuration trades slightly higher disk loading for overall reduction in system weight, resulting in a more efficient design for the given mission.
Overall, the sensitivity-based optimization demonstrates that careful tuning of rotor pitch, aspect ratio, and battery configuration can significantly improve quadrotor performance and reduce GTOW.

mapping my design → real components

My observation

Component →		Recommended
Propeller/rotor Radius ≈ 0.098 m Diameter ≈ 0.196 m (~7.7 inch) Chord ≈ 0.016 m Nb = 2 	☐ _____ ☐ _____ ☐ _____ ☐ _____	☐ 7x4.5 inch propeller (6045) ☐ Material: Nylon/ Carbon fiber Good efficiency at my RPM (~6000–9000)
Motor BLDC 	☐ Kv ≈ 766 RPM/V ☐ Current $\approx 4-6$ A ☐ Power $\approx 50-80$ W	2212 920KV motor OR 2212 1000KV motor Slightly higher KV → practical availability Works well with 3S battery Very standard for 7–8 inch props
Battery (LiPo) 	Cells = 3S Capacity $\approx 2670-3000$ mAh	☐ 3S 3000 mAh LiPo ☐ Discharge: $\geq 25C$ Slight margin for safety
ESC (Electronic Speed Controller) 	Current $\approx 5-6$ A	☐ 20A ESC (minimum) ☐ Prefer: 30A ESC (safe margin)
Airframe 	☐ GTOW $\approx 900-1100$ g ☐ Prop size $\approx 7-8$ inch	☐ F450 frame (450 mm) OR ☐ 500 mm carbon fiber frame